FIBRE DIMENSION PLUS TRANSCENDENCE DEGREE EQUALS THE TRANSCENDENCE DEGREE OF A FLAT LOCAL EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 14770-14814, immediately following the statement at 14754-14768.

Editorial note. Four cross-chapter pointers (Algebra, Morphisms) are recorded in assembly.external_references.

Editorial note. The two reduction steps (finite type model, valuation ring case) are both displayed inside the span.

Allowed standard prerequisites (editorial).

- Dimension of the fibre at a point for a flat local extension. Morphisms, lemma morphisms-lemma-dimension-fibre-at-a-point, cited in the proof. Identifies the fibre dimension with the dimension of the special fibre of the local extension.
- Dimension preserved under field extension. Algebra, lemma algebra-lemma-dimension-at-a-point-preserved-field-extension, cited in the proof. Keeps the fibre dimension unchanged when passing to a suitable localisation.
- Finite type domains over a valuation ring and their fibres. Algebra, lemma algebra-lemma-finite-type-domain-over-valuation-ring-dim-fibres, cited at proof lines 14811-14812. Gives the equality of dimension of special fibre and generic fibre in the valuation ring case that the proof reduces to.
- Dominating a local map by a finite type map of domains. Algebra, lemma algebra-lemma-dominate, cited in the proof. Replaces B by a domain C of finite type over A (with B a localisation of C), which is what allows the reduction to valuation rings.

Source reference. Original reference 00P4: focused-seeds/geometry/S9-morph-equality-dimensions/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00IA: focused-seeds/geometry/S9-morph-equality-dimensions/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00QK: focused-seeds/geometry/S9-morph-equality-dimensions/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 02FX: focused-seeds/geometry/S9-morph-equality-dimensions/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $A \to B$ be a local homomorphism of local rings. Assume*

(1) *A and B are domains and $A \subset B$,*
(2) *B is essentially of finite type over A, and*
(3) *B is flat over A.*

Then we have

$$\dim(B/\mathfrak{m}_A B) + trdeg_{\kappa(\mathfrak{m}_A)}(\kappa(\mathfrak{m}_B)) = trdeg_A(B).$$

Proof. Let C be an A-algebra of finite type and $\mathfrak{q}$ a prime of C such that $B = C_{\mathfrak{q}}$. We may assume C is a domain. We have $\dim_{\mathfrak{q}}(C/\mathfrak{m}_A C) = \dim(B/\mathfrak{m}_A B) + \mathrm{trdeg}_{\kappa(\mathfrak{m}_A)}(\kappa(\mathfrak{m}_B))$, see Morphisms, Lemma 02FX. Setting K equal to the fraction field of A we see by the same reference that $\dim(C \otimes_A K) = \mathrm{trdeg}_A(B)$. Thus we are really trying to prove that $\dim_{\mathfrak{q}}(C/\mathfrak{m}_A C) = \dim(C \otimes_A K)$. Choose a valuation ring A' in K dominating A, see Algebra, Lemma 00IA. Set $C' = C \otimes_A A'$. Choose a prime $\mathfrak{q}'$ of C' lying over $\mathfrak{q}$; such a prime exists because

$$C'/\mathfrak{m}_{A'}C' = C/\mathfrak{m}_A C \otimes_{\kappa(\mathfrak{m}_A)} \kappa(\mathfrak{m}_{A'})$$

which proves that $C/\mathfrak{m}_A C \to C'/\mathfrak{m}_{A'}C'$ is faithfully flat. This also proves that $\dim_{\mathfrak{q}}(C/\mathfrak{m}_A C) = \dim_{\mathfrak{q}'}(C'/\mathfrak{m}_{A'}C')$, see Algebra, Lemma 00P4. Note that $B' = C'_{\mathfrak{q}'}$ is a localization of $B \otimes_A A'$. Hence B' is flat over A'. The generic fibre $B' \otimes_{A'} K$ is a localization of $B \otimes_A K$. Hence B' is a domain. If we prove the lemma for $A' \subset B'$, then we get the equality $\dim_{\mathfrak{q}'}(C'/\mathfrak{m}_{A'}C') = \dim(C' \otimes_{A'} K)$ which implies the desired equality $\dim_{\mathfrak{q}}(C/\mathfrak{m}_A C) = \dim(C \otimes_A K)$ by what was said above. This reduces the lemma to the case where A is a valuation ring.

Let $A \subset B$ be as in the lemma with A a valuation ring. As before write $B = C_{\mathfrak{q}}$ for some domain C of finite type over A. By Algebra, Lemma 00QK we obtain $\dim(C/\mathfrak{m}_A C) = \dim(C \otimes_A K)$ and we win. □